
MODEL SELECTION AND VALIDATION REPORT (remove_5_biomass model)

OBSERVATIONS:1701

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_5_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	5.84	0.27
remove_5_tric	0.48	0.03
remove_5_teve	-0.46	0.03
sbt_mean	0.15	0.10
sbt_sd	-0.01	0.05
sst_sd	0.04	0.04
log_chla_mean	0.04	0.04
depth	0.11	0.05
log_sum_fishing	0.09	0.03

Dispersion parameter: 1.01

Spatiotemporal AR1 correlation (rho): 0.93

Matérn range: 311.58

Spatiotemporal marginal AR1 SD: 0.88

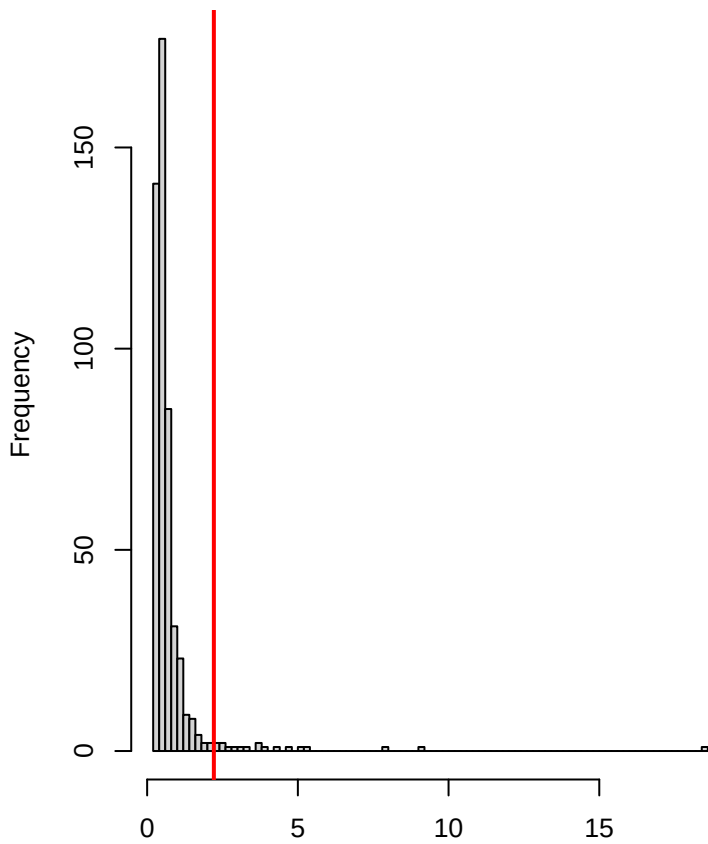
REML criterion at convergence: 11317.658

See ?tidy.sdmTMB to extract these values as a data frame.

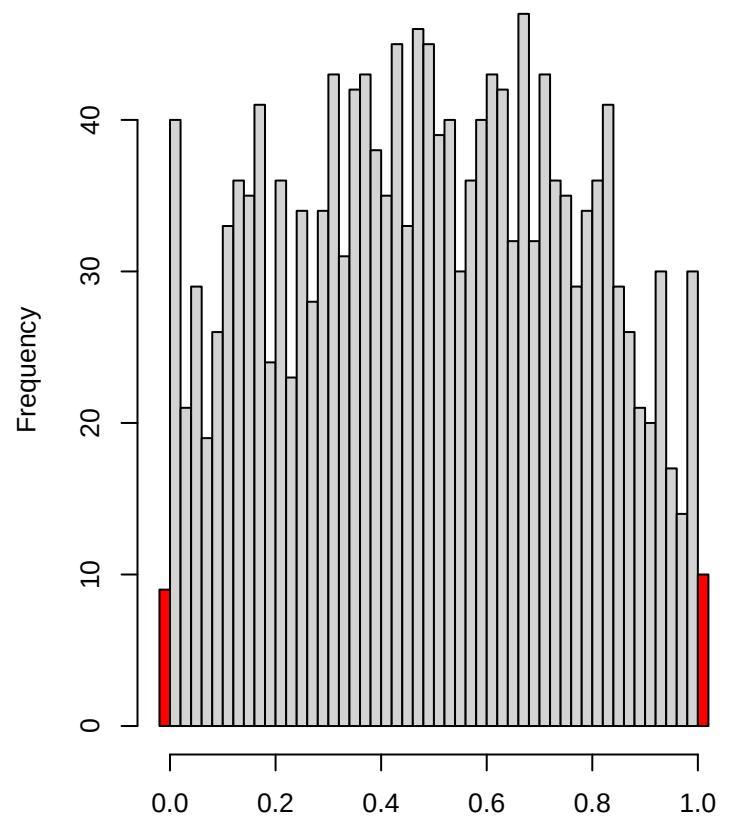
SELECTION STATISTICS

	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	22661.32	2.260754
spatiotemporal_random_fields_ar1_with_anisotropy	15	22663.58	114.661099
spatial_random_fields	12	22778.24	3.601694
spatial_random_fields_with_anisotropy	14	22781.84	14.380147
spatiotemporal_random_fields_iid	12	22796.22	2.341465
spatiotemporal_random_fields_iid_with_anisotropy	14	22798.56	152.985443
error_only	10	22951.55	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated

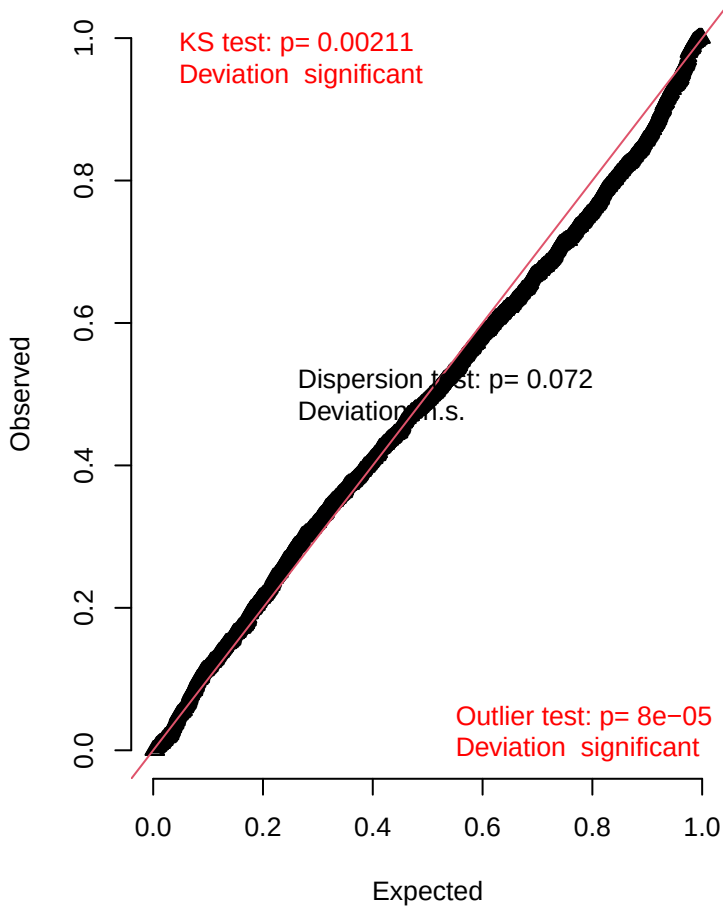


Outlier test significant



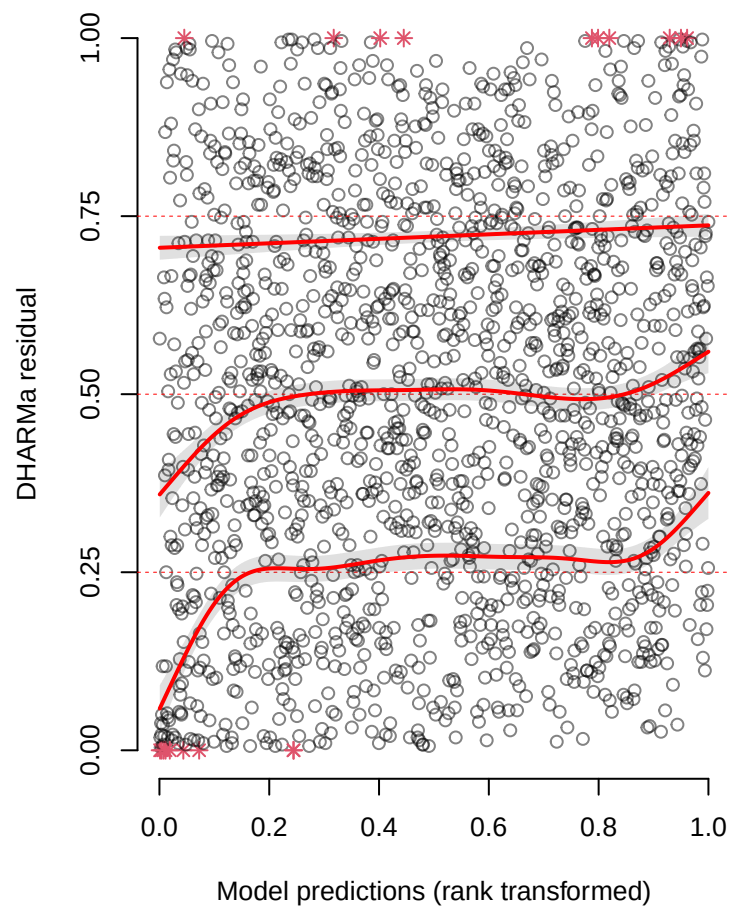
Simulated values, red line = fitted model. p-value (two.sided) =

QQ plot residuals

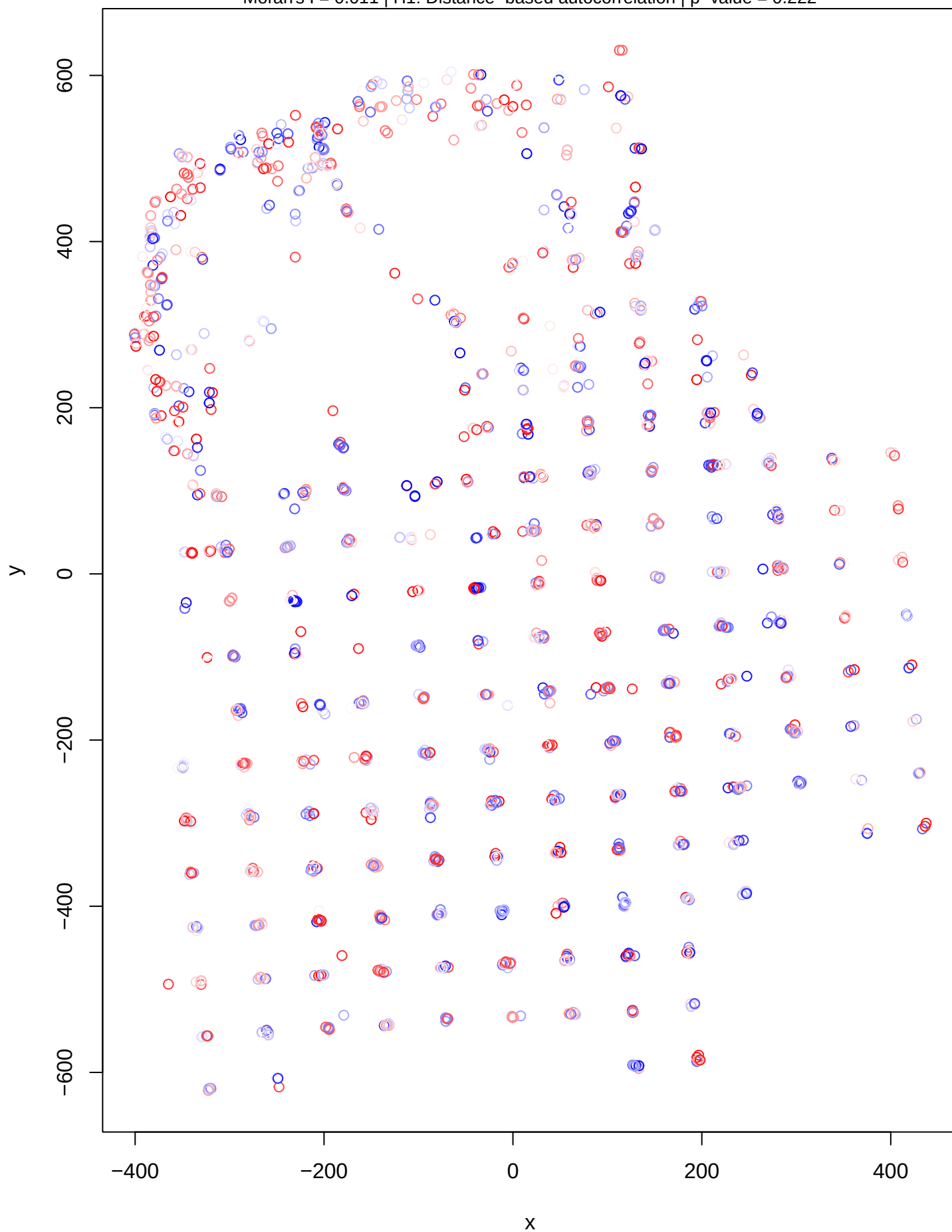


Residuals (outliers are marked red)

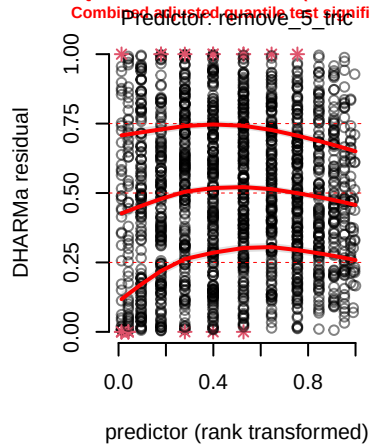
DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



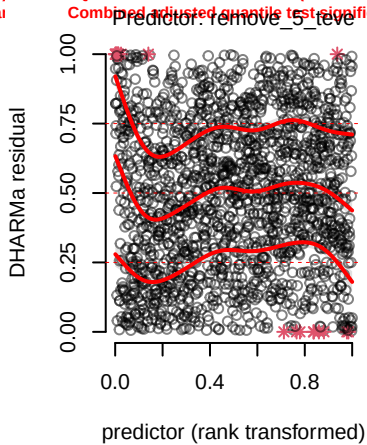
DHARMa Moran's I test for distance-based autocorrelation
Moran's I = 0.011 | H1: Distance-based autocorrelation | p-value = 0.222



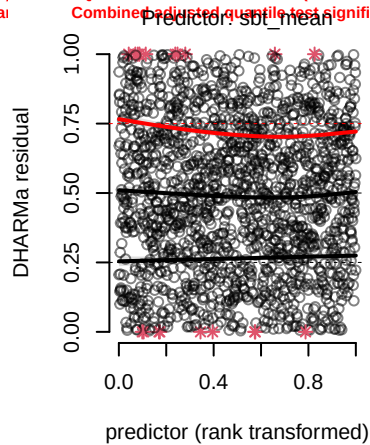
DHARMA residual Residual vs. predictor
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



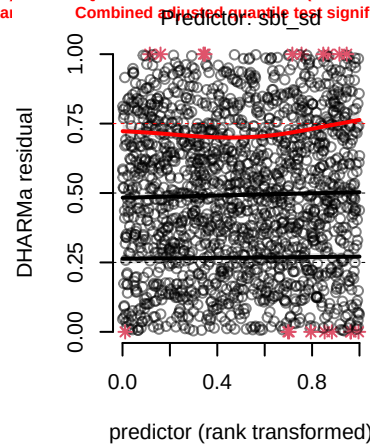
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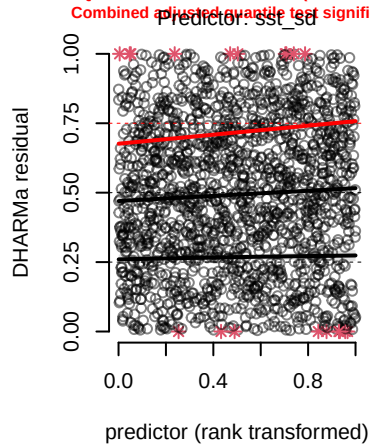
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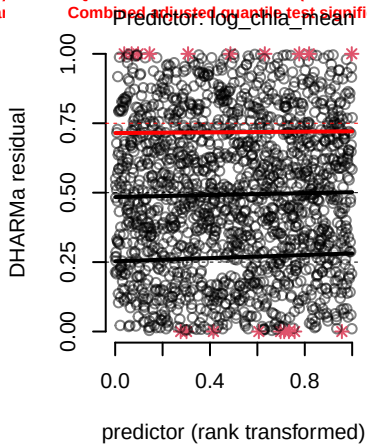
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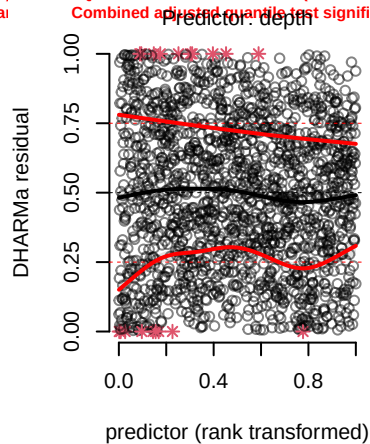
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